

JOANA OWUSU-APPIAH

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PROFILE

Medical imaging and computer vision engineer with an Erasmus Mundus Master's in Medical Imaging and Applications and hands-on R&D experience in surgical computer vision. Experienced in processing laparoscopic video and CT data, developing deep-learning pipelines for 3D pose estimation, detection and image registration, and evaluating image-guided surgery systems using patient and phantom data. Strong background in Python, PyTorch, OpenCV, medical image analysis, segmentation, registration, quantitative validation, and interdisciplinary collaboration with clinicians and engineers.

EDUCATION

Summer Geometry Initiative (SGI), Virtual Internship

Massachusetts Institute of Technology (MIT)

July 2025 – August 2025

Project: Worked on geometry processing and optimisation problems spanning robotic trajectory planning and medical image generation. Generated tetrahedral volume meshes from MRI data using SimNIBS and studied algorithms for handling non-manifold topology in medical datasets.

Erasmus Mundus Joint MSc Medical Imaging and Applications (MAIA)

Spain, France, Italy

September 2023 – September 2025

- Developed practical experience in medical image processing, image registration, segmentation, 3D reconstruction, quantitative imaging, and computational analysis across CT and MRI, together with deep learning for medical imaging and Software as a Medical Device (SaMD) development and validation.

BSc Biomedical Engineering

Kwame Nkrumah University of Science and Technology (KNUST), Ghana *September 2017 – November 2021*

Thesis: Exploring the use of the OpenFlexure Microscope in Ghanaian Hospitals for Routine Clinical Diagnosis.

SKILLS

Programming:	Python, C++, MATLAB, SQL, Linux, Git
Surgical Computer Vision:	Laparoscopic video, anatomical detection and segmentation, stereo endoscopy, 3D pose estimation, camera calibration, phantom validation
Deep Learning:	PyTorch, TensorFlow, Ultralytics YOLO, OpenCV, Scikit-learn, ONNX export and runtime, latency benchmarking
Medical Imaging:	CT, MRI, ASL perfusion MRI, DICOM, registration, segmentation, 3D anatomical processing, tetrahedral meshing
Tools:	3D Slicer, SimpleElastix, SimNIBS, ITK-SNAP, SPM, NiBabel, Nilearn, Fujifilm Synapse 3D, Meshroom
Evaluation:	Cross-validation, model validation, calibration, FROC, Target Registration Error, Mutual Information
Languages:	English (Native), French (B1)

PROFESSIONAL EXPERIENCE

Surgical Augmented Reality (SURGAR), France

MSc Thesis Research Intern – 3D4Kids Programme

February 2025 – June 2025

- Processed multimodal surgical data including video, images, 3D meshes, CT-derived anatomy, and numerical pose data to estimate intraoperative kidney position for image-guided surgical navigation.
- Developed and systematically benchmarked a 3D-to-2D registration pipeline from rectified stereo images using Siamese networks and feature fusion, evaluating the approach against an existing baseline and using the findings to redirect the methodology.
- Curated laparoscopic video data from seven patient procedures, extending annotations and preparing datasets for model development and quantitative evaluation.

- Extracted kidney vasculature from abdominal CT using 3D Slicer, Fujifilm Synapse 3D, and computer vision techniques including skeletonisation, supporting 3D anatomical modelling for surgical applications.
- Collaborated with clinicians and engineers during system demonstrations and an early-feasibility clinical investigation, using clinical feedback to refine the research workflow.

Danube Private University (Austria - Virtual)

Research Intern

August 2024 – November 2024

- Segmented DICOM images of soft-tissue phantoms, extracted quantitative features from defined regions of interest, and developed predictive models validated against known ground truth.

NASA, TOPS Nebula Cohort, Virtual

Researcher

March 2024 – April 2024

- Completed training in open-source scientific research and developed a data-management plan for medical imaging data, with emphasis on reproducibility, privacy, and responsible data sharing.

WamiAgro Limited, Ghana

Data Engineering Intern

June 2023 – August 2023

- Designed a data architecture for an AgriTech start-up covering data collection, optimised ETL paths using Apache Airflow, and data modelling and governance.
- Implemented a data strategy document that increased the company's yearly investment revenue by 30%. [Link to blog](#)

Department of Biomedical Engineering, KNUST, Kumasi-Ghana

Research and Teaching Assistant

September 2021 – September 2022

- Prepared scientific teaching materials and supported more than 1,000 undergraduate students in Basic Electronics, Bioinstrumentation, Research Methodology, and Arduino-based hardware programming.

COMPUTATIONAL MEDICAL IMAGING PROJECTS

Anatomical Structure Detection in Laparoscopic Cholecystectomy

[GitHub](#)

Tools: Python, PyTorch, Ultralytics YOLO26, ONNX Runtime, Endoscapes2023

- Fine-tuning YOLO26 on 1,186 annotated frames from 120 cholecystectomy procedures to localise five hepatocystic structures and surgical tools; selective-extraction and COCO-to-YOLO pipeline with programmatic verification of video-disjoint splits.
- Dataset characterisation of class imbalance and object-size distribution, and accuracy measured against inference latency across GPU PyTorch and CPU ONNX targets, at median and 95th percentile frame time.

Longitudinal CT Registration for COPD

Tools: Python, SimpleElastix, NumPy, Pandas

- Automated multi-stage registration of longitudinal lung CT using affine and non-rigid B-spline transformations to model anatomical change over time; quantitatively evaluated registration performance using Target Registration Error and Mutual Information.

Brain MRI Tissue Segmentation with SPM

Tools: SPM (MATLAB), Python

- Preprocessed and skull-stripped brain MRI NIfTI volumes and generated tissue probability maps for quantitative separation of brain tissues. [GitHub](#)

Preclinical Alzheimer's Detection Using ASL Perfusion MRI

Tools: Python, NiBabel, Nilearn, Scikit-learn, SciPy

- Quantified cerebral blood flow across 24 brain regions in amyloid-positive and amyloid-negative adults; evaluated regional differences statistically and developed a cross-validated classifier for amyloid status. [GitHub](#)

Breast Cancer AI - Clinical Validation Pipeline

Tools: Python, mammography & DBT datasets, HuggingFace / ONNX

- Built a reproducible validation pipeline for a black-box breast cancer detection model on 2D mammography and 3D digital breast tomosynthesis, evaluating discrimination, FROC localisation, calibration, screening

behaviour, and subgroup performance. [GitHub](#)

H&E Histopathology Tissue Classification with Explainable AI

Tools: Python, PyTorch, EfficientNet-B0, Grad-CAM

- Developed a 9-class colorectal tissue classifier using transfer learning and Grad-CAM, achieving 99.77% test accuracy and a quadratic weighted kappa of 0.9985. [GitHub](#)

Brain Tumour Segmentation & Classification

Tools: Python, classical machine learning, deep learning

- Applied medical image analysis, segmentation, classical machine learning, and deep learning methods to brain-tumour MRI. [GitHub](#)

Multi-class Skin Melanoma Classification - Computer Aided Diagnostic System

Tools: Python, PyTorch, OpenCV, Scikit-learn, NumPy, Pandas, CUDA

- Developed a deep learning pipeline for 3-class dermoscopic image classification (melanoma, basal cell carcinoma, squamous cell carcinoma), reaching an 85% kappa score with an ensemble of ResNet-50, Inception V3 and Swin Transformer V2. [GitHub](#)

SCIENTIFIC SOFTWARE & REPRODUCIBILITY

ArmKit - Scientific Documentation and Reproducibility Workflow

Tools: Git, GitHub Actions, Sphinx, MyST, reStructuredText

- Built a version-controlled scientific documentation workflow with automated API documentation, continuous integration, validation checks, and GitHub Pages publishing. [GitHub](#)

MindLink - Web Application

Tools: HTML, CSS, JavaScript, Java, Struts, Apache Tomcat, MySQL

- Mental health support platform for engineering students, supporting appointment scheduling with doctors, psychological tests with results, and emergency appointments. [GitHub](#)

CONFERENCES

- **EuroPython 2024 & 2025**, Prague – PR and Marketing Team (Organizer).
- **MAIA Day 2024 & 2025** – Presented MSc research at the 2025 edition to faculty, students, alumni, and clinical and industry partners.

ONLINE COURSES

- **Introduction to Medical Software** – Yale University (Coursera), 2026. Software as a Medical Device, clinical evaluation, risk classification, FDA, EU MDR, and IMDRF concepts.
- **Privacy Law, Data Protection & Regulatory Compliance** – University of Pennsylvania (Coursera), 2026. HIPAA, GDPR, and global data-protection frameworks.

TECHNICAL REPORTS

Master's Thesis

- Stereo-Based Landmark-Free Kidney Pose Estimation for AR-Guided Robot-Assisted Partial Nephrectomy. [Link](#)

Open Data Compilation - University of Barcelona

- An Overview of Open Data Sources for the Pan-African Research Infrastructure for Collaborative Biomedical Imaging and Artificial Intelligence (AFRICAI-RI) Projects. [Link](#)